

What is Regex?

Regex (Regular Expression) is used to:

- 👉 Find text
- 👉 Check text
- 👉 Extract text
- 👉 Replace text

inside a string.

Step 1: Import Module

```
import re
```

1 Search a Word

```
import re
```

```
text = "I love Python"
```

```
result = re.search("Python", text)
```

```
print(result)
```

If word exists → it shows match

If not → None

✓ 2 Match from Beginning

```
re.match("I", "I love Python") # Works
```

```
re.match("love", "I love Python") # Not works
```

`match()` checks only from start.

✓ 3 Find All Numbers

```
import re
```

```
text = "My numbers are 123 and 456"
```

```
numbers = re.findall("\d+", text)
```

```
print(numbers)
```

Output:

```
['123', '456']
```

`\d` means digit (0–9)

`+` means one or more

✓ 4 Replace Word

```
import re
```

```
text = "I like Java"
```

```
new_text = re.sub("Java", "Python", text)
```

```
print(new_text)
```

Important Basic Symbols

Symbol	Meaning
\d	Digit (0–9)
\w	Letter or number
\s	Space
.	Any character
+	One or more
*	Zero or more
^	Start of string

\$

End of string

✓ Example: Check Only Digits

```
import re
```

```
text = "12345"
```

```
if re.search("^\\d+$", text):
```

```
    print("Only numbers")
```

🧠 Very Simple Summary

- Import **re**
 - **search()** → find one
 - **findall()** → find all
 - **sub()** → replace
 - **\\d** → number
 - **+** → one or more
-